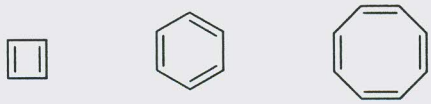
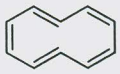
Organic Cume, March 14, 2014

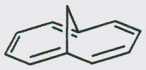
1. Discuss what is meant by "aromaticity" and "aromatic stabilization". For the latter term, graph with exact energy losses/gains and discuss.
2. By ^{1}H NMR, benzene showns a singlet absorption between 7-8 ppm. Why?
3. For the three compounds shown, discuss aromaticity and anti-aromaticity in terms of molecular orbital theory. In this discussion clearly show how to determine based on MO theory why a compound is anti-aromatic.



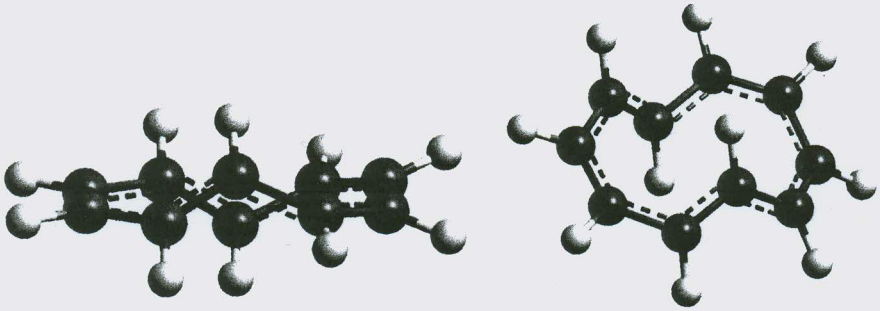
4. [10] Annulene is unstable and very reactive but 1,6-methanoannulene can be isolated and measurements indicate that it is aromatic. Ball and stick models are shown below and both are non-planar. Discuss the differences in stability between these two compounds in terms of the theory you presented above.



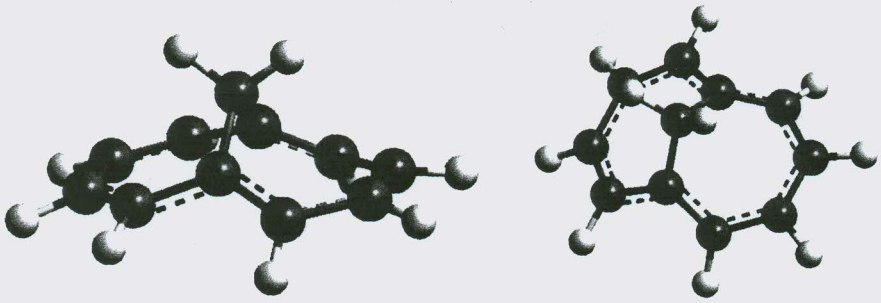
[10] annulene



1,6-methanoannulene



[10] annulene



1,6-methanoannulene